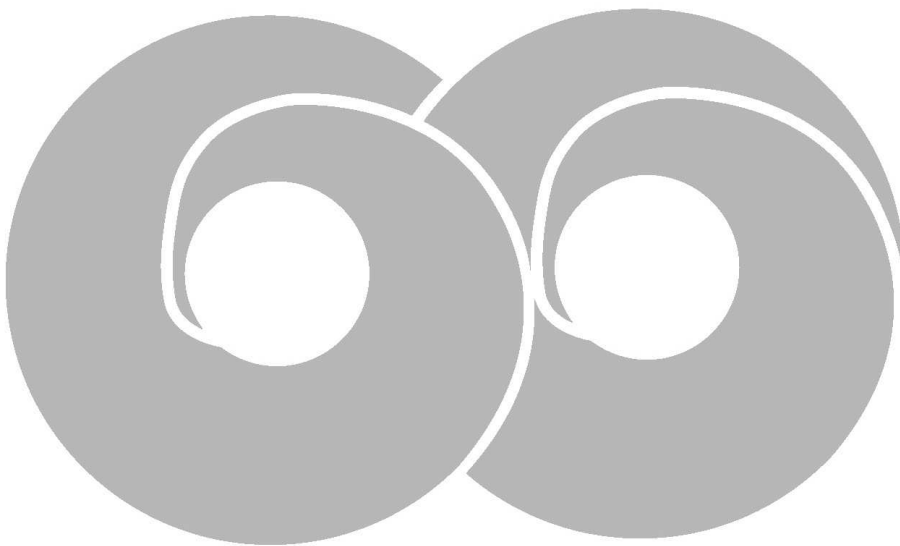




Installation and Maintenance Instructions



Screw Vacuum Pumps

COBRA NC 0630 C

with direct cooling water system



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0870567092 / 150116 / Original instructions / Modifications reserved

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Introduction

Congratulations on your purchase of the Busch vacuum pump. With watchful observation of the field's requirements, innovation and steady development Busch delivers modern vacuum and pressure solutions worldwide.

These operating instructions contain information for

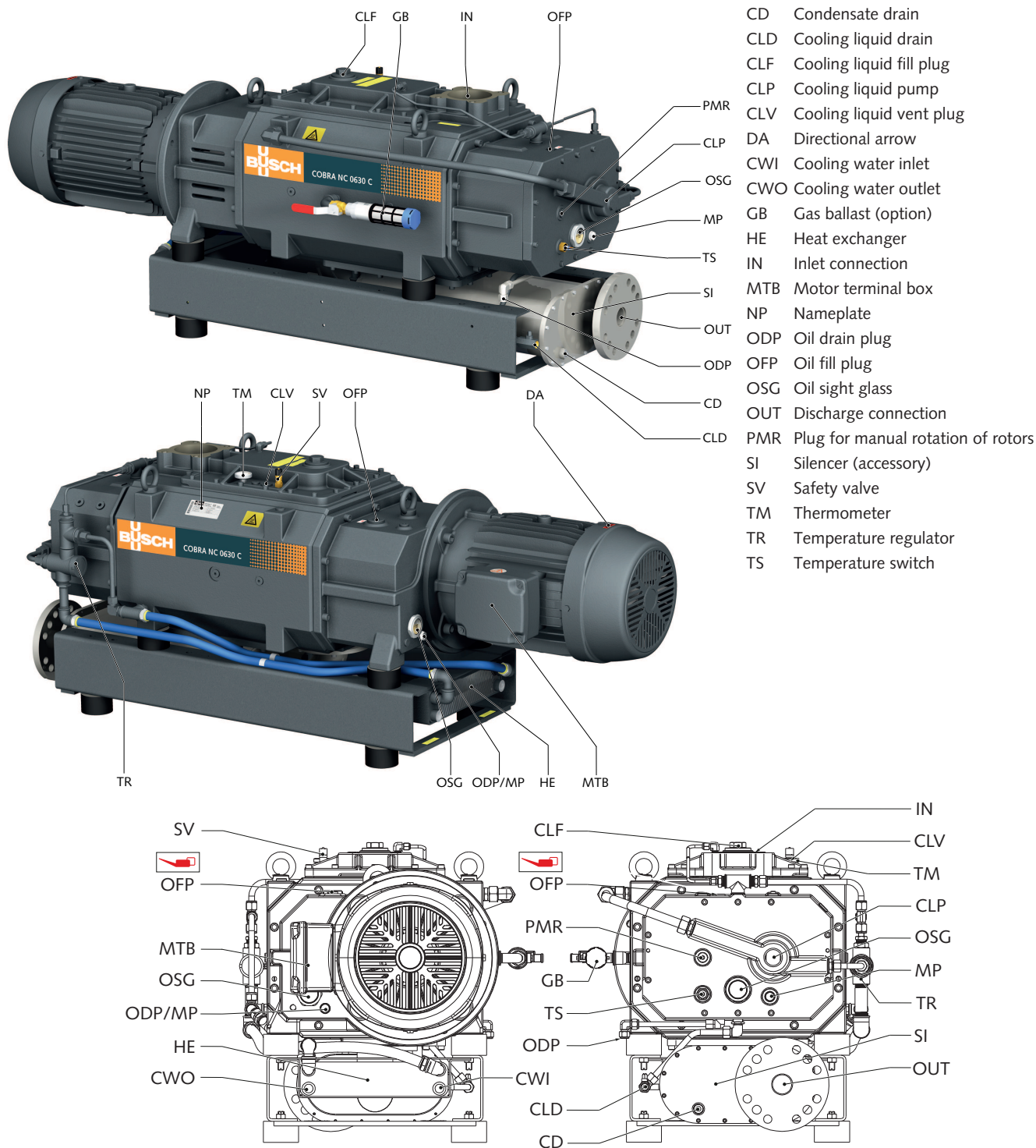
- product description,
- security,
- transport,
- storage,
- installation and commissioning,
- maintenance,
- overhaul,
- trouble shooting

of the vacuum pump.

For the purpose of these instructions, "handling" the vacuum pump means the transport, storage, installation, commissioning, influence on operating conditions, maintenance, troubleshooting and overhaul of the vacuum pump.

Prior to handling the vacuum system, these operating instructions shall be read and understood. If anything remains to be clarified please contact your Busch representative!

Keep these operating instructions and, if applicable, other pertinent operating instructions available on site.



Product Description

Use

The COBRA NC vacuum pumps are designed for use in the field of industrial applications and similar industries.

They can be used to draw gases and gas mixtures.



WARNING

When using toxic, inflammable and/or explosive gases, make sure that the system corresponds in design to applicable local and national safety regulations and that all applicable safety measures are followed.

All product-specific safety regulations must be observed.

Solid particles must not get into the vacuum pump. Procedural errors can result in the pump drawing in a certain quantity of liquid. If the pump has drawn in liquid, a short drying time will be necessary at the end of the process. In case a silencer (SI) is fitted at the outlet:

◆ Drain the silencer (CD)

The maximum admissible temperature of drawn gas depends on the inlet pressure and the nature of the gases. The lower the inlet pressure (Pa), the higher can be the drawn gas temperature (T_{Gas}). We can consider as air the following indications::

- Pa < 50 mbar, T_{Gas} < 200°C
- Pa > 50 mbar, T_{Gas} < 70°C

The vacuum pump is designed for use in an environment which is non-explosive.

Max. permissible number of starts per hour: 2

The vacuum pump is suitable for continuous operation at any inlet pressure between atmospheric pressure and ultimate pressure.

Principle of operation

The COBRA NC vacuum pumps are screw vacuum pumps with a cooling liquid circuit.

The COBRA NC vacuum pumps operate according to the principle of screw pumps. Two parallel screws rotate in opposite directions in the pump body. Entering gases are trapped between the pitches of the screws and the pump body. The gases are conveyed by the rotation of the screws to the exhaust side where they are discharged.

The COBRA NC vacuum pump is driven by an air-cooled motor.

Cooling

The vacuum pump is cooled by a cooling liquid (mix of water and glycol) inside the cooling liquid chambers of the cylinder and the inlet side end plate (B-side). This indirect circuit is done using an integrated recirculating pump (CLP).

The cooling liquid is cooled by a plate heat exchanger (HE) which must be connected up to the water main.

The cooling circuit is equipped with a temperature regulator (TR) fitted upstream of the plate heat exchanger (HE). When the cooling liquid temperature exceeds 55°C, the temperature regulator opens (mechanical opening) and allows the cooling liquid to get into the heat exchanger.

The cooling water inlet temperature must be checked.



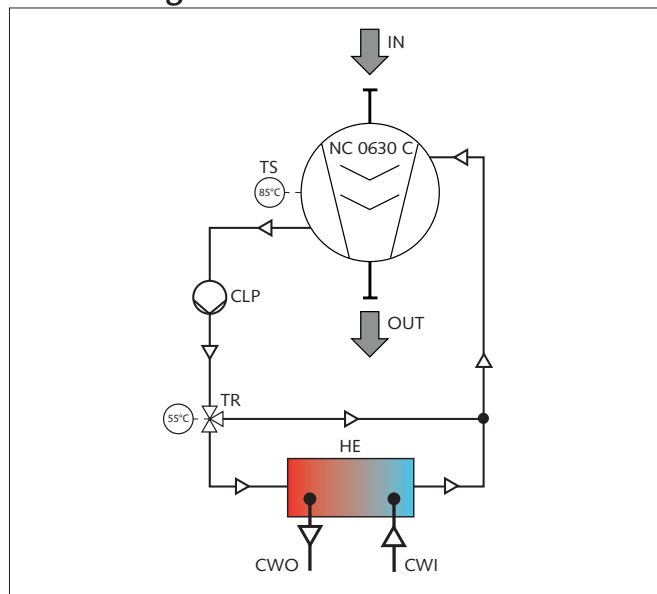
NOTE

Before first start-up of the vacuum pump, check the oil level and the cooling liquid level, correct if needed.

Do not forget to connect the cooling water supply before the first start-up.

Operation without oil, cooling liquid and cooling water can result in damage to the vacuum pump.

Process diagram



Monitoring of the vacuum pump

A temperature switch (TS) monitors the oil temperature. When the oil temperature exceeds 85°C, 30 seconds after the warning signal goes off, the pump must be switched off.

Sealing systems

The COBRA NC vacuum pumps are equipped with labyrinth seals on the motor side (A-side) and inlet side (B-side) as standard. Other sealing systems are available as an option:

NC 0630 C	Motor side (A)	Inlet side (B)
Standard	Labyrinth	Labyrinth
Option	Oil lubricated mechanical seal	Oil lubricated mechanical seal
	Oil lubricated mechanical seal	Labyrinth

Options/ Accessories

A resistance thermometer PT100 can be provided instead of temperature switch (TS)

A silencer (SI) at the outlet of the pump reduces the noise of the pump.

A gas ballast for ambient air (GB). The gas ballast increases the pump capacity to handle condensable vapours that may be present in the process stream.

The gas ballast has an influence on the ultimate pressure of the pump (see table below).

Operating frequency	Ultimate pressure	
	with closed gasballast	with opened gasballast
50 Hz	0.01 hPa	0.05 hPa
60 Hz	0.01 hPa	0.05 hPa

A barrier gas system allows the supply of compressed air or nitrogen to the sealing system. This option seals off the process gases and the oil. The barrier gas is fed into the cylinder intermediary chambers on the motor side.

A nitrogen panel fitted to the base frame allows the supply of nitrogen to a number of different points on the vacuum pump. The system allows to adjust pressure and volume flow separately. This gas can be used in a number of different ways:

- A purge gas system fitted to the inlet flange allows to flush the vacuum pump after use or during operation. This system consists of a solenoid valve which enables to open and close the purge circuit. The filtered gas is fed directly into the inlet flange.
- Barrier gas for the labyrinth seals or the oil-lubricated single mechanical seal: this option seals off the process gases and the gear oil. The nitrogen is fed into the intermediary chambers.
- Dilution gas: this option prevents the formation of condensates or dilutes them, depending on the application. The nitrogen is fed directly into the cylinder of the vacuum pump.

A flushing device allows the vacuum pump to be flushed with an appropriate liquid for the process type

On/ Off switch

The vacuum pump is delivered without on / off switch. The control of the vacuum pump must be provided in the course of the installation.

Safety

Intended use

DEFINITION: For the purpose of correct understanding, the “handling” of the vacuum pump implies the transport, storage, installation, commissioning, the influence on operating conditions, maintenance, troubleshooting and overhaul of the vacuum pump.

The vacuum pump is intended for industrial use. It should only be handled by qualified staff.

The different applications for use and operational limits of the vacuum pump as laid out in the “Product Description” and the “Installation Prerequisites” of the vacuum pump must be observed both by the manufacturer of the machinery into which the vacuum pump is to be incorporated and by the end user.

The need for personal safety regulations depends mainly on the application the pump will be used in. The end user must provide the operators with all necessary means and tools and must inform his personnel about any dangers emanating from the processed products.

The operator of the vacuum pump must observe the safety regulations and must train and instruct his personnel accordingly.

Local regulations regarding the motors and electric control elements must be observed when installing the pump in potentially explosive environments.

The maintenance instructions must be observed and respected.

It is vital that these installation and maintenance instructions are read and understood before the vacuum pump is used. If you have any doubts, please contact your local Busch representative.

Safety information

The vacuum pump has been designed and manufactured in accordance with the latest technical and safety standards. Nevertheless, residual risks may remain.

A lot of safety information is mentioned in these Installation and Operating Instructions as well as on the pump. The safety instructions must be observed. The safety information can quickly be detected through key words like DANGER, WARNING and CAUTION and is defined as follows:



DANGER

Disregard of this safety note will always lead to accidents with potentially fatal injuries and serious damages.



WARNING

Disregard of this safety note may lead to accidents with potentially fatal injuries and serious damages.



CAUTION

Disregard of this safety note will always lead to accidents with minor injuries and damages to property.

Noise emission

For the sound pressure level in free field according to EN ISO 2151, see “Technical Data”.



CAUTION

The vacuum pump emits noise of high intensity.

Risk of damage to the hearing.

Persons staying in the vicinity of a non noise insulated vacuum pump over extended periods shall wear ear protection.

Transport

The COBRA NC vacuum pumps are tested and checked in our factory before careful packing. Check the packaging for transport damage when the goods arrive. The pump can withstand temperatures between -25°C and +55°C during transport.

Transportation of packaged pump

Packed on a pallet, the vacuum pump can be transported with a forklift.

Transport in unpacked state

The vacuum pump is bolted to a pallet or a base plate:

- ◆ Remove the bolts between the vacuum pump and the pallet or base plate



CAUTION

Do not walk, work or stand under suspended loads.



CAUTION

Please check out the weight of the vacuum pump before lifting it up (see “Technical Data”).

Use adequate lifting gear for this.

NOTE: The eyebolts are fitted more or less at equal distance from the centre of gravity of the vacuum pump incl. drive motor. If any accessories that could upset the balance of the vacuum pump, are installed, or if the vacuum pump is delivered without drive motor, it is necessary to add a belt or rope at a suitable point when lifting the pump.

- Attach lifting gear securely to the eye bolts
- Use lifting gear with a crane hook equipped with safety latch.
- Lift the vacuum pump

In case the vacuum pump was bolted to a pallet with fixing bolts:

- ◆ Unscrew the fixing bolts in the base frame



CAUTION

Once filled with oil, the vacuum pump should not be lifted anymore.

Prior to every transport make sure that the oil has been drained from the vacuum pump

The packaging material must be disposed of in accordance with local and national regulations.

Storage

Short-term Storage

- Make sure that the inlet/ discharge connections are closed
- Store the vacuum pump
 - if possible in its original packaging,
 - indoors,
 - dry,
 - in a dust free room and
 - free from vibrations

Removal from storage

Before starting up a vacuum pump that has been stored outside the building for a while, the vacuum pump must be moved to a room with ambient temperature, where it should rest for a day.

Conservation

In case of adverse ambient conditions (e.g. aggressive atmosphere, frequent temperature changes) conserve the vacuum pump immediately.

In case of favourable ambient conditions conserve the vacuum pump if a storage of more than 3 months is anticipated.

- Drain oil, cooling liquid and cooling water
- Make sure that all openings are hermetically sealed; use adhesive tape to fasten loose parts (such as o-rings, flat seals, etc.).
- Wrap the vacuum pump in VCI film
- Store the vacuum pump
 - if possible in its original packaging,
 - indoors,
 - dry,
 - in a dust free and
 - vibration free area

NOTE: VCI is the abbreviation for “volatile corrosion inhibitor”. The VCI molecule is an organic corrosion inhibitor in the vapour phase. Integrated in various carriers such as film, cardboard, paper, foam, liquid and powder, it protects parts against corrosion as a result of its action in vapour phase. However, VCI packaging can attack plastic surfaces and surfaces of other elastomers. If in doubt, please contact your nearest distributor. VCI packaging provides several years of protection against corrosion, even under harshest conditions: overseas shipment, extended storage before use.

Starting-up of the vacuum pump after storage

- Please ensure that all protective agents such as gaskets, plugs or adhesive tapes that were used for the protection of the pump, are removed.
- Commission the vacuum pump as described in chapter “Installation and Commissioning”

Installation and Commissioning

Installation prerequisites



CAUTION

In case of non-compliance with the installation prerequisites, particularly in case of insufficient cooling:

Risk of damage or destruction of the vacuum pump and its components!

Risk of personal injury!

The installation prerequisites must be complied with.

- Please ensure that the integration of the vacuum pump is compliant with the safety requirements of the Machine Directive 2006/42/EC (concerning the responsibility of the system's manufacturer into which the vacuum pump is to be incorporated, please also refer to the note in the EC-Declaration of Conformity).



WARNING

Local regulations regarding the motors and electric control elements must be observed when installing the pump in potentially explosive environments. Before start-up, make sure that all safety measures have been followed.

Local installation

- Make sure that the environment of the vacuum pump is not potentially explosive
- Make sure that the following ambient conditions are adhered to:
 - Temperature: +5 - +50 °C
 - Pressure: atmospheric
 - Humidity range: 20 to 95%
 - Altitude: up to 1000 m
- Make sure that the cooling water fulfills the following requirements:
 - Temperature: +10 - +30 °C
 - Pressure: 1 - 6 bar (relative), differential pressure between cooling water inlet and outlet ≥ 1 bar
 - Flow: 12 l/min
 - Hardness: $< 5^\circ$ dGH

NOTE: 1° (german degree = 1° dGH) = $1,78^\circ$ (french degree) = $1,25^\circ$ (english degree) = 17,9 mg/kg CaCO₃ (american hardness)

- Make sure that the cooling water is neutral and clean
- Make sure that the environmental conditions correspond to the protection class of the motor (according to motor nameplate)
- Make sure that the vacuum pump is placed on or fastened to a horizontal surface
- Make sure that the vacuum pump is level and even
- Make sure that the vacuum pump cannot inadvertently or intentionally be used as a support for heavy objects
- Make sure that the vacuum pump cannot be hit by falling objects
- Make sure that the vacuum pump is easily accessible and that the selected installation site fulfills the requirements for assembly / dismantling
- Make sure that no temperature-sensitive part (such as plastic, wood, cardboard, paper, electronic parts) come into direct contact with the hot surfaces of the vacuum pump
- Make sure that the installation site or assembly area is ventilated in such a way that adequate cooling of the vacuum pump is assured



CAUTION

During operation the surface of the vacuum pump may reach a temperatures of 90° C.

Risk of burns!

- Make sure that the vacuum pump cannot be touched inadvertently during operation, provide a guard if necessary
- Make sure that the oil sight glasses (OSG) will remain easily accessible

If the oil change is meant to be performed on site:

- ◆ Make sure that the oil drain plugs as well as the oil filler plugs, are easily accessible.

Suction Connection

- Make sure that the protection that prevents the ingress of foreign matter during transport, has been removed before connecting up the vacuum pump to the piping.



CAUTION

Do not put hands into the inlet aperture.

Risk of body damage !



CAUTION

The ingress of foreign objects or liquids can destroy the vacuum pump.

In case the inlet gases can contain dust or other foreign solid particles:

- ◆ Make sure that a suitable filter or protection screen is installed at the inlet of the vacuum pump
- Make sure that the nominal diameter of the inlet line is at least equal to the diameter of the inlet flange to prevent a drop in the performance of the vacuum pump in the case of a smaller cross-section.
- Make sure that the vacuum pump is connected with leak proof lines.



CAUTION

Once the inlet line has been connected up, make sure that the system does not leak. Leakage of dangerous substances must be prevented!

- Make sure that the inlet line is equipped with a shut-off device upstream of the inlet flange, so that the flow of drawn gases can be stopped
- Make sure that the inlet line does not exercise any load on the inlet flange. Use bellows if necessary.
- The inlet flange (IN) has the following dimension:
 - DN100 ISO-K, DIN 28404

Discharge connection



CAUTION

Do not put hands into the outlet aperture.

Risk of body damage !

The following instructions for connection to the discharge side only apply if the drawn gas is discharged into a suitable environment by the vacuum pump.

- Make sure that the protection, that was fitted to prevent the ingress of particles during transport, has been removed before the vacuum pump is connected up to the vacuum pipe
- Make sure that the nominal diameter of the discharge line corresponds at least to the diameter of the exhaust flange of the vacuum pump in order to prevent an increased counter pressure at the discharge
- Make sure that the discharge line is fitted in such a way so that it prevents any liquids or condensates re-entering the vacuum pump (discharge line sloping away from the pump)
- Make sure that the discharge line does not exercise any load on the exhaust flange. Use bellows if necessary.
- The outlet flange (OUT) has the following dimension:
 - ◆ Pump outlet
 - DN100 ISO-K, DIN 28404
 - ◆ Silencer outlet (SI) (accessory)
 - DN80 PN16, EN 1092-1
 - ASME B16.5 3" 150 lbs
- Maximal counter pressure at the discharge:
 - 0,2 bar

Cooling water connections

The cooling water is generally connected up with a flexible hose.

Connection dimension:

- cooling water inlet (CWI): G 1/2, ISO 228-1
- cooling water outlet (CWO): G 1/2, ISO 228-1

Nitrogen system connection (option)

The connection of the nitrogen panel (NIP) is generally done using flexible hoses (diameter 1/4")

Connection dimension:

- G 1/4, ISO 228-1

Electrical connection/ Checks

- Make sure that the requirements according to EMC-Directive 2004/108/EC as well as the current EN-standards, electrical and occupational safety directives and the local or national regulations respectively, are complied with (this is the responsibility of the designer of the machinery into which the vacuum pump is to be incorporated; see also the corresponding comments in the EC-Declaration of Conformity).
- Make sure that the power supply is compatible with the data on the nameplate of the drive motor
- Make sure that an overload protection according to EN 60204-1 is provided for the drive motor
- Make sure that the drive of the vacuum pump will not be affected by electric or electromagnetic interference; if unsure please seek advice from your Busch representative

Installation

- Make sure that the "Installation Prerequisites" are complied with
- Fit or install the vacuum pump at its final location

Electrical connection



WARNING

Risk of electrical shock, risk of damage to equipment.

Electrical installation work must only be executed by qualified personnel that knows and observes the following regulations:

- IEC 364 or CENELEC HD 384 or DIN VDE 0100, respectively,
- IEC-Report 664 or DIN VDE 0110,
- BGV A2 (VBG 4) or corresponding national accident prevention regulation.



CAUTION

The connection schemes given below are typical. Specific orders or deviating connection schemes for certain markets may apply.

Risk of damage to the drive motor!

Please check drive motor connections inside the terminal box and refer to the drive motor connection instructions.



CAUTION

Providing the motor and the installation allow for it, the operating frequency of the motor must not go below 20Hz, which corresponds to a nominal motor rotational speed of 1200 rpm.

Risk of damage to the vacuum pump!



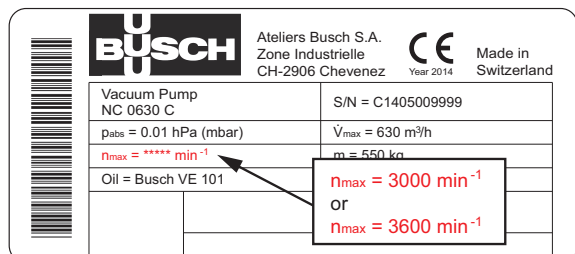
WARNING

The operating rotational speed of the drive motor must correspond to the pump's nameplate value. Do not exceed the mentioned rotational speed.

$n_{\max} = 3000 \text{ min}^{-1} \rightarrow 50 \text{ Hz}$

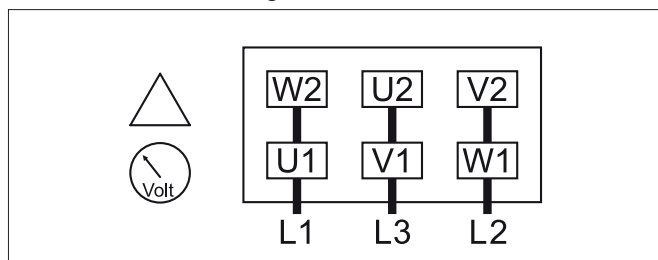
$n_{\max} = 3600 \text{ min}^{-1} \rightarrow 60 \text{ Hz}$

Risk of damage to the vacuum pump!

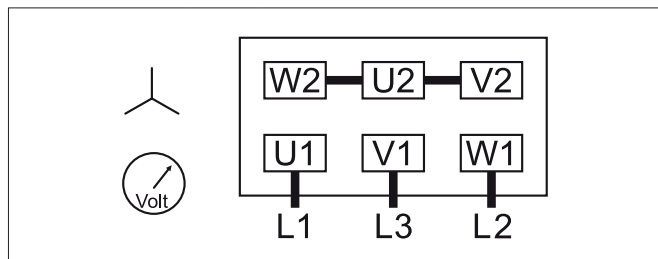


- Connect the drive motor electrically
- Connect the earth line

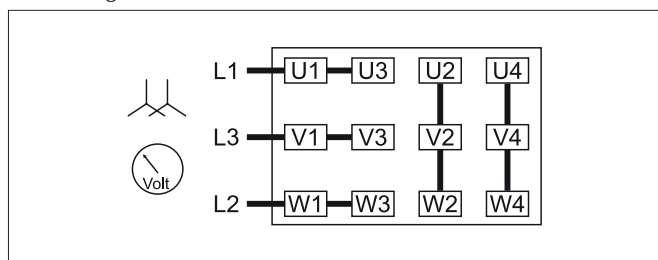
Delta connection (Low voltage):



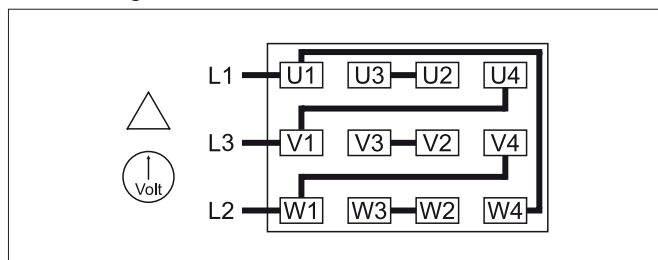
Star connection (High voltage):



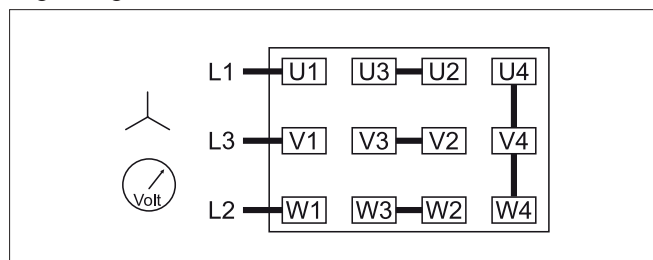
Double star connection, multi-voltage motor with 12 terminals (Low voltage):



Delta connection, multi-voltage motor with 12 terminals (Middle voltage):



Star connection, multi-voltage motor with 12 terminals (High voltage):



CAUTION

Operation of the pump in the wrong direction of rotation, even for a short period of time, can destroy the vacuum pump.

Risk of damage to the vacuum pump!

Prior to starting-up of the vacuum pump, please ensure that the vacuum pump is connected up correctly.

- Determine the direction of rotation with the arrow sign (DA) on the fan cover of the motor
- The motor fan should rotate in clockwise direction
- Press the start button and hold briefly
- Watch the fan of the drive motor and determine the direction of rotation just before the fan stops

If the fan is turning in the wrong direction:

- ◆ Switch around any two of the drive motor wires in the terminal box (MTB)

Pump temperature monitoring connection

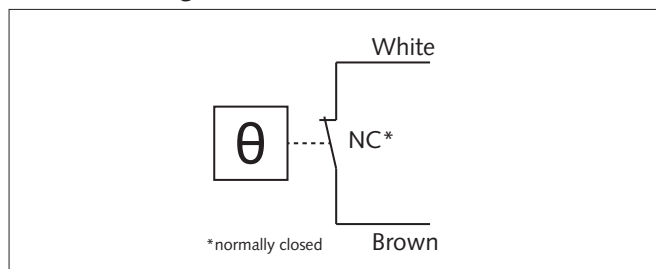
The pump temperature is monitored by a temperature switch (TS).

The electrical wiring of the temperature switch has to be done in order that the vacuum pump stops when the temperature switch trips (break contact).

Technical data

Technical data	
Set value	85 °C ±5 °C
Switching cycles	10'000
AC current cosφ = 1	2,5 A / 250 VAC
AC current cosφ = 0,6	1,6 A / 250 VAC
DC current	1,25 A / 48 VDC
Resistance contact max.	50 mΩ
Bounce time contact max.	1 ms
High voltage max.	2 kV
Cable operating temperature max.	motionless -30°C ... +80°C
Cable operating temperature max.	moved -5°C ... +70°C

Electrical wiring



Barrier gas flow meter connection (option)

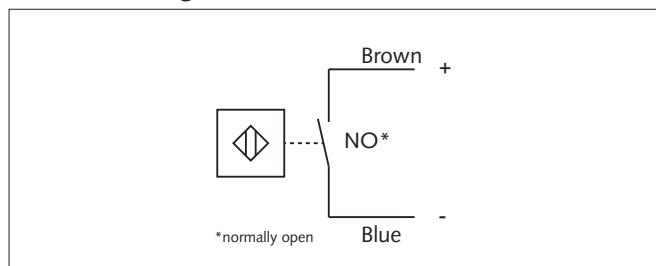
A flow meter with contact fitted to the nitrogen panel enables the monitoring of the gas volume flow which is fed into the vacuum pump.

The flow meter must be connected in such a way that the vacuum pump switches off when the volume flow drops below the min. set flow value.

Technical data

Technical data	Flow meter
Type	Orobiche 212-J-1-G1-C1
Control range	0,75 - 5,75 SLM
Pressure	3 bar
Inductive proximity switch type	Pepperl+Fuchs RC15-14-N3
Switching element function	NAMUR, NO
Current consumption	1 - 3 mA

Electrical wiring



Solenoid valve connection (option)

Opening/ closing of the nitrogen barrier gas circuit

A solenoid valve connected to the barrier gas circuit enables to control the opening/ closing of the nitrogen barrier gas circuit.

The nitrogen barrier gas circuit depends on the application and is defined by the end user of the vacuum pump.

Opening/ closing the nitrogen purge gas circuit

A solenoid valve, connected to the inlet flange, enables to control the opening/ closing of the nitrogen purge gas circuit.

The nitrogen purge gas circuit depends on the application and is defined by the end user of the vacuum pump.

Opening/ closing the nitrogen dilution gas circuit

A solenoid valve connected to the dilution gas circuit enables to control the opening/ closing of the nitrogen dilution gas circuit.

The dilution gas circuit depends on the application and is defined by the end user of the vacuum pump.

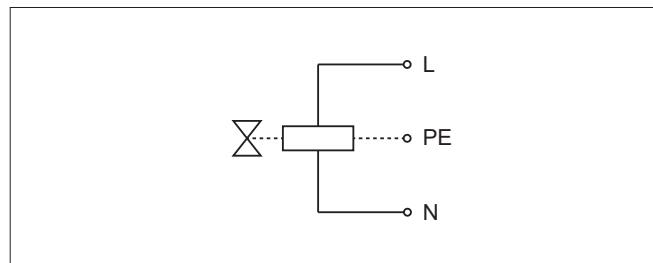
Opening/ closing the flushing liquid device

A solenoid valve connected to the flushing liquid device enables to control the opening/ closing of the flushing liquid device.

Technical data

Technical data	Solenoid valve
Nominal voltage	24 VDC
Ambient temperature	-40 ... +65 °C
Current	360 mA
Power	8 W

Electrical wiring



Level switches connection of the flushing liquid device (option)

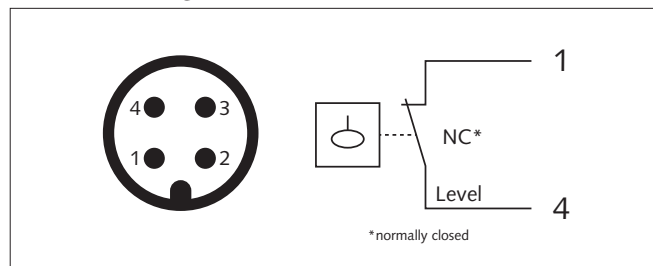
The level switches of the flushing liquid device monitor two different levels in the vessel of the flushing liquid.

The upper level switch is used for an early warning signal and the lower level switch is used for an alarm signal.

Technical data

Technical data	Level switch
Switching element function	NAMUR, NC
Current consumption	0.6 - 3.5 mA
Power	6 mW - 38 mW
Electrical connection	M12x1; 4-pin
Ambient temperature	-50 - +70°C

Electrical wiring



Standard wiring colours of supplied connection cable

Brown	1
White	2
Blue	3
Black	4

Connecting up of lines/ pipes

- Connect the suction line
- Connect the discharge line
- Make sure that all provided covers, guards, hoods etc. are fitted

Filling up with oil

The COBRA NC vacuum pumps are delivered without oil (see chapter "Oil Type" for information on recommended oils).

- Prepare the quantity of oil specified in the table "Oil quantity"

NOTE: The quantity of oil specified in the operating instructions is of informative nature only. Check the oil level with the help of the two oil sight glasses (OSG) on the vacuum pump.

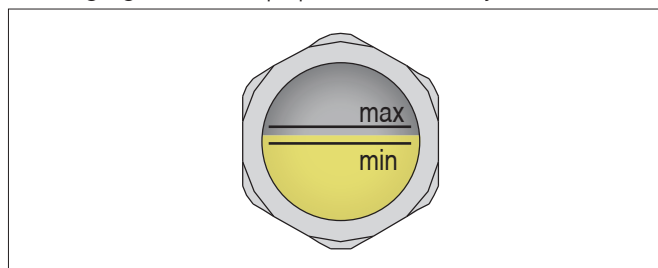
- Unscrew the oil filler plugs (OFP)
- Fill in oil
- Make sure that the filling level is between the Min and Max of the oil sight glass (OSG)
- Make sure that the seals in the oil filler plugs are not damaged, replace plugs if necessary
- Fit the oil filler plugs and tighten up
- Switch on the vacuum pump

In case the suction line is equipped with a shut-off valve:

- ◆ Close the shut-off valve

In case the suction line is not equipped with a shut-off valve:

- ◆ Cover the inlet connection with a rubber mat
- Let the vacuum pump run for a few minutes
- Switch off the vacuum pump and wait for a few minutes
- Make sure that the filling level is between the Min and Max of the oil sight glass (OSG). Top-up with oil if necessary



In case the suction line is equipped with a shut-off valve:

- ◆ Open the shut-off valve

In case the suction line is not equipped with a shut-off valve:

- ◆ Remove the rubber mat from the suction flange and connect the suction line to the suction flange
- Before any transport, make sure that the oil has been drained out of the vacuum pump.



CAUTION

Once filled with oil, the vacuum pump should not be lifted or moved anymore.



CAUTION

The vacuum pump must remain in a horizontal position once it has been filled with oil.

Filling in cooling liquid

The COBRA NC vacuum pumps are delivered without cooling liquid (see chapter "Cooling liquid types" for information on the recommended cooling liquids).

- Prepare the quantity of cooling liquid as specified in the table "Cooling liquid quantity"

NOTE: The quantity of cooling liquid specified in the installation handbook is of informative nature only. Follow the procedure of filling the cooling liquid.

- Remove the cooling liquid filler plug (CLF) on cylinder upper plate
- Remove the cooling liquid vent plug (CLV) on cylinder upper plate

- Fill in cooling liquid by filler hole until the cooling liquid runs out of the cooling vent plug (CLV) on cylinder upper plate
- Refit the cooling liquid vent plug (CLV) on cylinder upper plate
- Fit the cooling liquid filler plug (CLF) on cylinder upper plate
- If there is any spillage on the surface of the vacuum pump, wipe it up
- Start the vacuum pump

If the inlet line is equipped with a shut-off device:

- ◆ Close the shut-off device

If the inlet line is not equipped with a shut-off device:

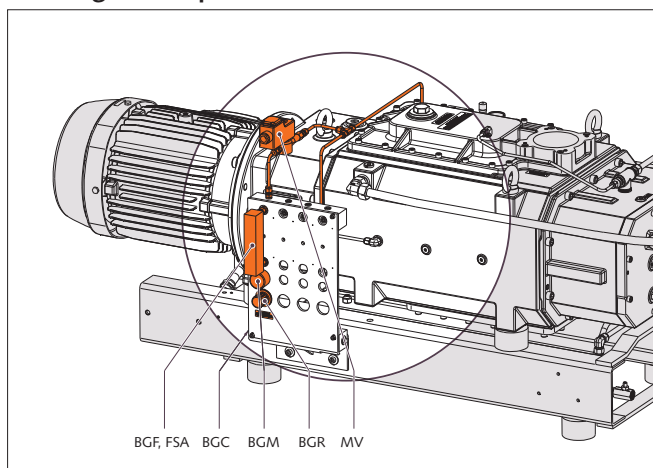
- ◆ Place a rubber plate on the suction flange
- Let the vacuum pump run for maximum 5 minutes
- Stop the vacuum pump and wait for a few minutes
- Remove the cooling liquid vent plug (CLV) on cylinder upper plate
- Check that the filling level is just under the cylinder upper plate

In case the cooling liquid level is below the required level:

- ◆ Fill in more cooling liquid
- Refit the cooling liquid vent plug (CLV) on cylinder upper plate

Barrier gas system (option)

Barrier gas with panel



BGC	Barrier gas connection G 1/4, ISO 228-1
BGF, FSA	Barrier gas flow meter with flow switch
BGM	Barrier gas manometer
BGR	Barrier gas pressure regulator
MV	Solenoid valve

Set values:

Type	Volume flow [SLM]*	bar [relative]
NC 0630 C	4 ... 5	3

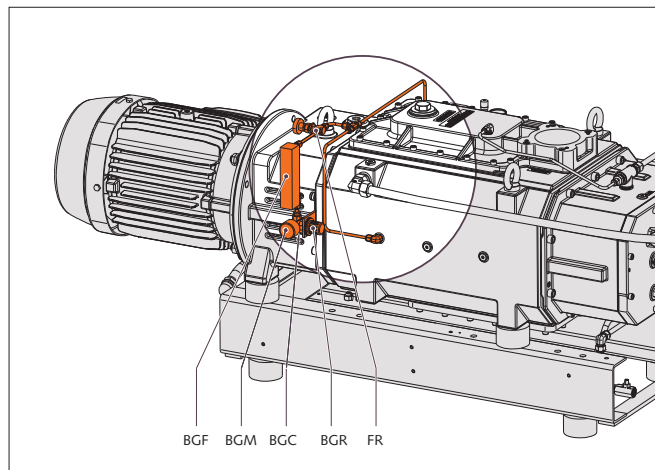
Adjustment of the barrier gas:

- Connect the different supervisory instruments (flow switch (FSA) of the flow meter (BGF) and the solenoid valve (MV))
- Connect the gas main supply up to the connection on the control panel (BGC)
 - ◆ Gas type: nitrogen or compressed air
- Adjust the barrier gas pressure with the help of the pressure regulator (BGR) to 3 bar (relative)
 - ◆ The manometer (BGM) shows a visual display of the barrier gas pressure

NOTE: The barrier gas volume flow is approx. 4 SLM. The barrier gas volume flow should not drop below the set flow value.

* SLM = Standard liter per minute

Barrier gas without panel



BGC	Barrier gas connection G 1/4, ISO 228-1
BGF	Barrier gas flow meter
BGM	Barrier gas manometer
BGR	Barrier gas pressure regulator
FR	Flow regulator

Set values:

Type	Volume flow [SLM]*	bar [relative]
NC 0630 C	5 ... 5,75	3

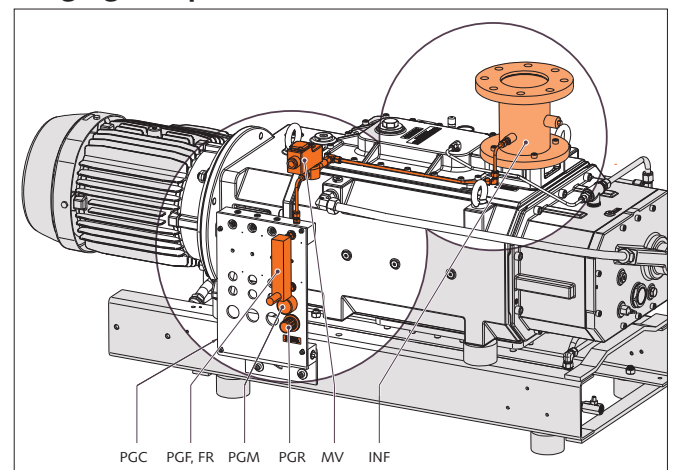
Adjustment of the barrier gas:

- Connect the gas main supply up to the connection on the control panel (BGC)
 - ◆ Gas type: nitrogen or compressed air
- Adjust the barrier gas pressure with the help of the pressure regulator (BGR) to 3 bar (relative)
 - ◆ The manometer (BGM) shows a visual display of the barrier gas pressure
- Adjust barrier gas volume flow with the help of the flow regulator (FR) from 5 to 5,75 SLM
 - ◆ The flow meter (BGF) shows a visual display of the barrier gas volume flow

NOTE: The barrier gas volume flow should not drop below the set flow value.

* SLM = Standard liter per minute

Purge gas (option)



PGC	Purge gas connection G 1/4, ISO 228-1
PGF, FR	Purge gas flow meter with flow regulator
PGM	Purge gas manometer
PGR	Purge gas pressure regulator
INF	Inlet flange
MV	Solenoid valve

Set values:

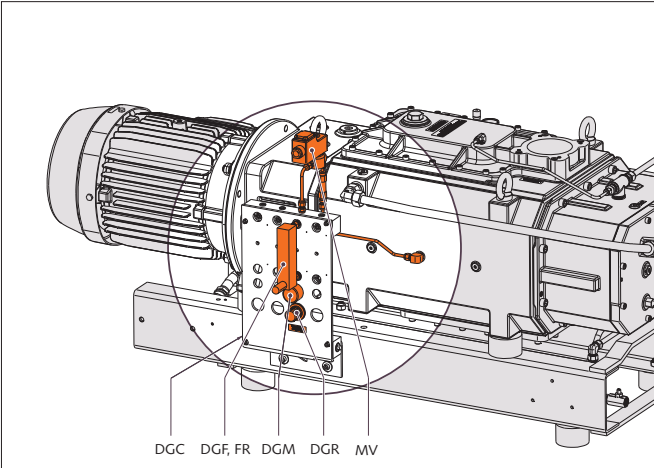
Type	Volume flow [SLM]*	bar [relative]
NC 0630 C	≥ 100	2,5

Adjustment of the purge gas:

- Connect the solenoid valve (MV)
- Connect the gas main supply up to the connection on the control panel (PGC)
- Switch on the vacuum pump, the inlet is closed
 - ◆ Open the solenoid valve (MV) of the purge gas circuit
- Adjust the purge gas pressure with the help of the pressure regulator (PGR) to 2,5 bar (relative)
 - ◆ The manometer (PGM) shows a visual display of the barrier gas pressure
- Adjust the purge gas volume flow with the help of the flow regulator (FR) of the flow meter (PGF)
 - ◆ The purge gas volume flow must correspond to the set flow value
- Close the solenoid valve (MV) of the purge gas circuit

* SLM = Standard liter per minute

Dilution gas (option)



DGC	Dilution gas connection G 1/4, ISO 228-1
DGF, FR	Dilution gas flow meter with flow regulator
DGM	Dilution gas manometer
DGR	Dilution gas pressure regulator
MV	Solenoid valve

Set values

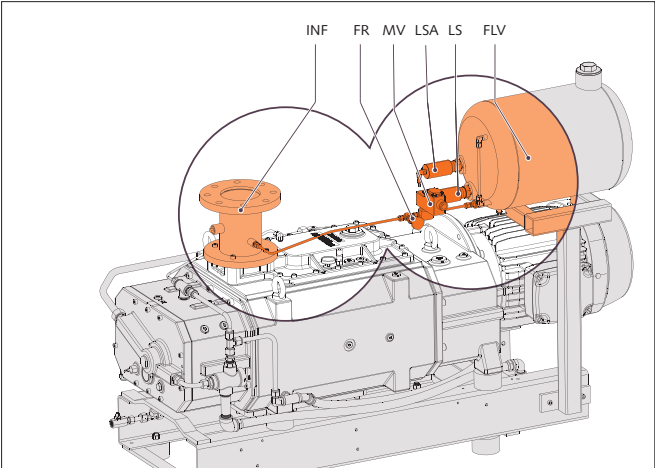
Type	Volume flow [SLM]*	bar [relative]
NC 0630 C	30	2,5

Adjustment of the dilution gas:

- Connect the solenoid valve (MV)
- Connect the gas main supply up to the connection on control panel (DGC)
- Switch on the vacuum pump, the inlet is closed
- Open the solenoid valve (MV) of the dilution gas circuit
- Adjust the dilution gas pressure with the help of the pressure regulator (DGR) to 2,5 bar (relative)
 - ◆ The manometer (DGM) shows a visual display of the dilution gas pressure
- Adjust the dilution gas volume flow with the help of the flow regulator (FR) of the flow meter (DGF)
 - ◆ The dilution gas volume flow must correspond to the set flow value
 - ◆ The dilution gas volume flow depends also on the application
- Close the solenoid valve (MV) of the dilution gas circuit
- Switch off the vacuum pump

* SLM = Standard liter per minute

Flushing liquid device (option)



FLV	Flushing liquid vessel
FR	Flow regulator
LSA/LS	Level switch
INF	Inlet flange
MV	Solenoid valve

Set values

Type	Liquid consumption [l/min]	motor speed [Hz]
NC 0630 C	1,0*	20

Adjustment of the flushing liquid device:

- Connect the solenoid valve (MV) and the two level switches (LSA/LS)
- Fill the flushing liquid vessel with a process compatible flushing liquid
- Switch on the COBRA NC vacuum pump with a reduced motor speed, the inlet is closed
 - ◆ motor speed: 20 Hz
- Open the solenoid valve (MV) of the flushing liquid circuit
 - ◆ It is possible to adjust the flushing liquid flow with the help of the flow regulator (FR)
- Close the solenoid valve (MV) of the flushing liquid circuit

NOTE: The duration and the flow rate of the flushing depend on the application

* These values can change depending on the application

Saving the operating parameters

As soon as the vacuum pump has been working for 30 minutes under normal conditions after having been switched on:

- Measure the working current of the motor and keep it as reference value for all future maintenance and repair work

Recommendations on operation

Application



WARNING

The vacuum pump is designed for operation under the conditions described below.

In case of disrespect of the above, risk of damage or destruction of the vacuum pump!

The COBRA NC vacuum pumps have been designed for use in industrial applications and similar industries.

They can be used to draw gases and gas mixtures.



WARNING

When using toxic, inflammable and/ or explosive gases, make sure that the system corresponds to applicable local and national safety regulations and that all applicable safety measures are followed. All product-specific safety regulations must be observed.

Solid particles must not enter the vacuum pump. In case of process errors, the pump can draw in a certain amount of liquids. If the pump has drawn in liquid, a short drying run will be necessary at the end of the process and, in case a silencer (SI) is mounted at the outlet:

- ◆ Drain the silencer (CD)

The vacuum pump is designed for use in an environment which is non-explosive.

Max. permissible number of starts per hour: 2

The vacuum pump is suitable for continuous use at any inlet pressure between atmospheric pressure and ultimate pressure.



CAUTION

During operation the surface of the vacuum pump may reach a temperatures of 90° C.

Risk of burns!

The vacuum pump must be protected against contact during operation. If touching the pump is unavoidable, wait until the surface temperature has cooled down or wear protective gloves.



CAUTION

The sound level of the pump within a certain perimeter of the pump is high.

Risk of hearing damage.

Users, who are spending a longer period of time in the vicinity of a non-insulated vacuum pump, must wear suitable ear protection.



CAUTION

The COBRA NC vacuum pumps are always delivered without oil.

Operation without oil will destroy the vacuum pump within a short period of time.

The vacuum pump must remain in a horizontal position once it has been filled with oil.



CAUTION

The COBRA NC vacuum pumps are always delivered without cooling liquid.

Operation without cooling liquid will destroy the vacuum pump within a short period of time!

- Make sure that all provided covers, guards, hoods etc. remain fitted
- Make sure that protective devices will not be disconnected
- Make sure that there is no leakage in the system, the escape of dangerous substances must be avoided
- Make sure that the "Installation Prerequisites" are complied with and will remain so, and ensure that adequate cooling is guaranteed

If the pump is shut down for a longer period of time:



CAUTION

If there is a risk of freezing, all the cooling water must be drained out of the vacuum pump if the pump is shut down for a longer period of time !

- Drain the cooling liquid
 - ◆ Put a drain tray underneath the cooling liquid drain valve (CLD)
 - ◆ Remove the cooling liquid vent plug (CLV)
 - ◆ Open the cooling liquid drain valve (CLD)
 - ◆ Drain the cooling liquid
 - ◆ Refit the cooling liquid vent plug (CLV)
 - ◆ Close the cooling liquid drain valve (CLD)
 - ◆ Collect the cooling liquid and re-use it or dispose of it according to local or national regulations
- Drain the cooling water
 - ◆ Disconnect the cooling water inlet / outlet connections (CWI, CWO)
 - ◆ Drain the cooling water completely
 - ◆ Drain the cooling water with the help of compressed air in order to prevent any risk of freezing or corrosion

NOTE: When the pump has not been in operation for a few days or when a sticky substance has been drawn, it is possible that the two rotor screws of the COBRA NC vacuum pump stick to each other. Unscrew the access plug (PMR) to the rotor screws. Loosen the rotors from each other with an Allen key and an extension by turning them clockwise by hand.

Switching the vacuum pump on/ off

First start-up of the system

- Make sure that the "Installation Prerequisites" are followed
- Open the cooling water supply

If the vacuum pump is equipped with a barrier gas system:

- ◆ Open the solenoid valve
- ◆ Open the gas supply

- ◆ Adjust the barrier gas pressure

If the vacuum pump is equipped with a purge gas system:

- ◆ Open the solenoid valve
- ◆ Open the gas supply
- ◆ Adjust the pressure and volume flow for the purge gas

If the vacuum pump is equipped with a dilution gas system:

- ◆ Open the solenoid valve
- ◆ Open the gas supply
- ◆ Adjust the pressure and volume flow for the dilution gas

- Start the vacuum pump

If the vacuum pump is equipped with a solenoid valve at the inlet:

- ◆ Open the solenoid valve

If the vacuum pump is equipped with a shut-off valve at the inlet:

- ◆ Open the shut-off valve

Switching off the system

If the vacuum pump is equipped with a solenoid valve at the inlet:

- ◆ Close the solenoid valve

If the vacuum pump is equipped with a shut-off valve at the inlet:

- ◆ Close the shut-off valve

If the vacuum pump is equipped with a purge gas system:

- ◆ Open the solenoid valve on the purge gas system
- ◆ Flush the vacuum pump for 20 - 40 minutes
- ◆ Close the solenoid valve on the purge gas system

- Switch off the vacuum pump
- Close the cooling water supply

If the vacuum pump is equipped with a barrier gas, purge or dilution gas system:

- ◆ Close the gas supply
- ◆ Close the solenoid valve(s)

Maintenance



In case the vacuum pump has conveyed gases that have been contaminated with foreign materials that are dangerous to health, the oil and condensates will also be contaminated.

These foreign materials can infiltrate the pores, recesses and other internal spaces of the vacuum pump.

Danger to health when the vacuum pump is dismantled.

Danger to the environment.

Always wear protective clothing when carrying out maintenance work.



Only authorised personnel may carry out any dismantling on the vacuum pump. Before work begins, the operator of the vacuum pump must fill in a form or a "Declaration of Decontamination" that provides information on possible dangers and appropriate measures.

If this form has not been filled in completely and signed by a responsible person, the vacuum pump may not be dismantled.



Before any maintenance work is started, a safety perimeter of at least 1m around the machine must be set up.



During operation the surface of the vacuum pump may reach a temperatures of 90° C.

Risk of burns !



The oil temperature can reach a value of 100°C!

Danger of burns!

- Leave the vacuum pump to cool down prior to any action that requires touching the vacuum pump
- Make sure that the oil and the cooling water have been drained before moving the vacuum pump
- Prior to disconnecting inlet and outlet pipes, make sure that all piping is vented to atmospheric pressure

Maintenance program

NOTE: The maintenance intervals depend on the individual operating conditions. The intervals given below should be considered as initial guidelines which should be shortened or extended as appropriate. In particularly heavy duty operation such as high dust loads in the environment or in the process gas, it can become necessary to shorten the maintenance intervals significantly.

Monthly:

- Check the oil level (see "Checking the Oil")
- Check the level of the cooling liquid (see "Checking the Cooling liquid")
- Check the temperature of the cooling water (see "Checking the Cooling water")
- Check the vacuum pump for oil leaks - in case of leaks, have the vacuum pump repaired (Busch service)
- Check the vacuum pump for cooling liquid leaks - in case of leaks, have the vacuum pump repaired (Busch service)
- Check the vacuum pump for cooling water leaks - in case of leaks, have the vacuum pump repaired (Busch service)

Yearly:

In case of operation in a dusty environment:

- ◆ Make sure that the operating room is clean and free of dust; clean the room if necessary
- Make sure that the vacuum pump has been switched off and that it cannot accidentally be switched on again
- Check the electrical connections
- Carry out a visual inspection of the vacuum pump
- Check the operation of the gas ballast assembly (GB) (option), disassemble and clean as required. In the case of excessive discolouration of the inlet filter change this item

If the inlet is equipped with a mesh screen:

- ◆ Check the mesh screen at the inlet and clean it if necessary
- Check the correct operation of the measurement and safety equipment

Every 1000 operating hours:

If the discharge is equipped with a silencer (SI) (accessory):

- ◆ Bleed the condensation of the silencer through the purge system (CD)

Every 5'000 operating hours:

- Change the oil (see "Changing the Oil")
- Change the cooling liquid (see "Changing the cooling liquid")

If an filter is installed on the cooling water line:

- Check the filter, clean or replace it, if necessary

Every 10'000 operating hours:

- Check the seals and replace them if necessary
- Check the inlet and discharge lines and clean or replace them if necessary

If the vacuum pump is equipped with oil lubricated mechanical seals:

- Check the external look of the oil lubricated mechanical seals. If they are worn or damaged, replace them.

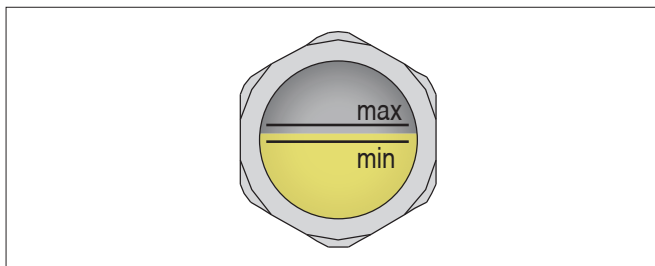
Every 16'000 operating hours, at the latest after 4 Years:

Have a major overhaul done on the vacuum pump (Busch service)

Checking the oil

Checking the oil level

- Check the oil level on the two oil sight glasses. This can be done during operation of the pump or during standstill
- Make sure that the filling level is between the Min and Max of the oil sight glass (OSG)



If the oil level is below the required level:

- ◆ Top up with oil (see "Topping up Oil")

If the oil level is above the required level:

- ◆ Check the evacuation of the condensates at the pump exhaust
- ◆ Drain the oil (see "Draining oil")

Topping up with oil

NOTE: Under normal conditions there should be no need to top up with oil in-between the recommended oil change intervals. A significant drop in the oil level indicates a malfunction (see "Troubleshooting").



CAUTION

Risk of injury (burns) with open oil filler orifice.

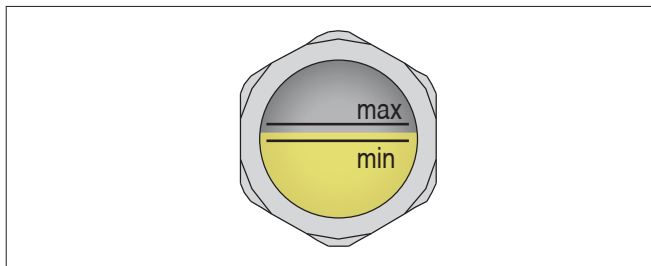
Risk of injury in case of badly screwed-in plugs.

Remove the oil filler plugs only when the vacuum pump is stopped.

The vacuum pump must only be operated when the oil filler plugs are firmly tightened up and do not leak.

- Make sure that the vacuum pump is switched off and cannot accidentally be switched on again
- Remove the oil filler plugs (OFP)

- Top up with oil
- Make sure that the filling level is between the Min and Max of the two oil sight glasses (OSG)



- Make sure that the seals of the oil filler plugs are not damaged and replace plugs if necessary
- Refit the oil filler plugs and tighten up

Life-span of the Oil

The oil life depends very much on the operating conditions. Under normal conditions the oil must be changed every 5000 operating hours or after 12 months, whichever comes first.

Under very unfavourable operating conditions the oil life can be less than 500 operating hours. A very short oil life time indicates either a malfunction (see "Troubleshooting") or unsuitable operating conditions.

If there is no experience available regarding the oil life under the prevailing operation conditions, it is recommended to have the oil analysed every 500 operating hours and thus to establish the oil change interval accordingly.

Changing the oil



WARNING

In case the vacuum pump has conveyed gases that have been contaminated with harmful foreign materials, the oil will also be contaminated.

Danger to health during the change of contaminated oil.

Danger to the environment.

Wear protective equipment during the change of contaminated oil.

Contaminated oil is hazardous waste and must be disposed of separately in compliance with applicable regulations.

Draining oil

NOTE: After switching off the vacuum pump at normal operating temperature wait no more than 20 minutes before the oil is drained.

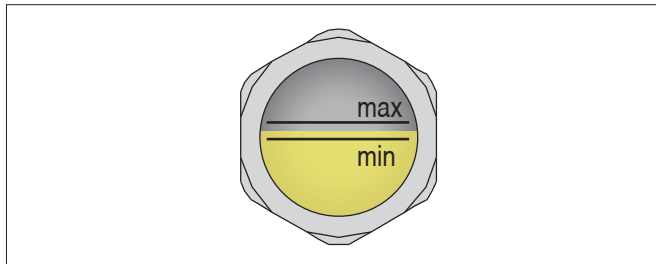
- Make sure that the vacuum pump is switched off and cannot accidentally be switched on again
- Make sure that the vacuum pump is vented to atmospheric pressure
- Put a drain tray underneath the oil drain plugs (ODP)
- Remove the oil drain plugs (ODP)
- Drain the oil
- Because of wear and tear on the seals replace the current drain plugs with new ones
- Remove the two magnetic plugs (MP), clean or change it if necessary
- Refit the two magnetic plugs (MP)
- Refit the oil drain plugs (ODP) and tighten up
- Dispose of the used oil in compliance with applicable regulations

Filling in new oil

- Prepare the quantity of oil needed (see "Oil type/quantity")

NOTE: The quantity of oil specified in the operating instructions is of informative nature only. Check the oil level with the help of the various oil sight glasses on the vacuum pump.

- Make sure that the oil drain plugs have been fitted properly and that they don't leak
- Remove the oil filler plugs (OFP)
- Fill in oil
- Make sure that the filling level is between the Min and Max of the two oil sight glasses (OSG)



- Make sure that the seals of the oil filler plugs are not damaged and replace plugs if necessary
- Refit the oil filler plugs (OFP) and tighten up

Checking the cooling liquid

Checking the level of the cooling liquid

- Make sure that the vacuum pump is switched off and cannot accidentally be switched on again
- Check the level of the cooling liquid
- Remove the cooling liquid vent plug (CLV) on cylinder upper plate
 - ◆ The filling level must be just under the cylinder upper plate

If the level is below:

- ◆ Top up with cooling liquid (see "Refilling cooling liquid")

Topping up with cooling liquid

NOTE: Under normal conditions there should be no need to top up with cooling liquid in-between the recommended change intervals. A significant drop in the cooling liquid level indicates a malfunction (see "Troubleshooting").

- Make sure that the vacuum pump is switched off and cannot accidentally be switched on again



CAUTION

Wait until pump has cooled down to ambient temperature

- Remove the cooling liquid filler plug (CLF) on the cylinder
- Remove the cooling liquid vent plug (CLV) on cylinder upper plate
- Fill in cooling liquid by filler hole until the cooling liquid runs out of the cooling vent plug (CLV) on cylinder upper plate
- Refit the cooling liquid vent plug (CLV) on cylinder upper plate
- Fit the cooling liquid filler plug (CLF) on cylinder upper plate
- Make sure that the seal rings of the cooling liquid filler plug and cooling liquid drain plugs are not damaged, replace plugs if necessary
- If there is any spillage on the surface of the vacuum pump, wipe it up

Changing the cooling liquid

Draining the cooling liquid

- Make sure that the vacuum pump is switched off and cannot accidentally be switched on again



CAUTION

Wait until pump has cooled down to ambient temperature

- Make sure that the vacuum pump is vented to atmospheric pressure
- Put a drain tray underneath the cooling liquid drain valve (CLD)
- Remove the cooling liquid vent plug (CLV)
- Open the cooling liquid drain valve (CLD)
- Drain the cooling liquid
- Refit the cooling liquid vent plug (CLV) and tighten up
- Close the cooling liquid drain valve (CLD)
- Collect the cooling liquid and re-use it or dispose of it according to local or national regulations

Filling up with new cooling liquid

- Prepare the quantity of cooling liquid needed (see "Cooling liquid type / quantity")

NOTE: The quantity of cooling liquid specified in the operating instructions is of informative nature only.

- Remove the cooling liquid filler plug (CLF) on cylinder upper plate
- Remove the cooling liquid vent plug (CLV) on cylinder upper plate
- Fill in cooling liquid by filler hole until the cooling liquid runs out of the cooling liquid vent plug (CLV) on cylinder upper plate
- Refit the cooling liquid vent plug (CLV) on cylinder upper plate
- Fit the cooling liquid filler plug (CLF) on cylinder upper plate
- If there is any spillage on the surface of the vacuum pump, wipe it up
- Start the vacuum pump

If the inlet line is equipped with a shut-off device:

- ◆ Close the shut-off device

If the inlet line is not equipped with a shut-off device:

- ◆ Place a rubber plate on the suction flange
- Let the vacuum pump run for maximum 5 minutes
- Stop the vacuum pump and wait for a few minutes
- Remove the cooling liquid vent plug (CLV) on cylinder upper plate
- Check that the filling level is just under the cylinder upper plate

In case the cooling liquid level is below the required level:

- ◆ Fill in more cooling liquid
- Refit the cooling liquid vent plug (CLV) on cylinder upper plate

Checking the cooling water

Checking the cooling water temperature

- Check regularly the cooling water temperature
 - ◆ Make sure that the specifications of the cooling water are followed

Checking the current consumption

- Check the current of the motor

An increased current indicates a fault (see "Troubleshooting")

Checking the silencer (SI) (accessory)

- Make sure that the condensates do not collect at the outlet of the vacuum pump
- Drain the condensates via the drain provided (CD) and collect them in a container
- Dispose of the condensates in compliance with applicable environmental protection regulations
- Check regularly the silencer and clean it if necessary



CAUTION

Wear protective clothing when carrying out maintenance work on the silencer.

Risk of contaminated process residues.

Checking the gas ballast (GB) (option)

- Check the operation of the gas ballast assembly, disassemble and clean as required. In the case of excessive discolouration of the inlet filter change this item

Overhaul



CAUTION

In order to achieve best efficiency and a long life the vacuum pump was assembled and adjusted with precisely defined tolerances.

This adjustment will be lost during dismantling of the vacuum pump.

It is therefore strictly recommended that any dismantling of the vacuum pump that is beyond of what is described in this manual shall be done by Busch service.



DANGER

In case the vacuum pump has conveyed gases that have been contaminated with foreign materials that are dangerous to health, the oil and condensates will also be contaminated.

These foreign materials can infiltrate the pores, recesses and other internal spaces of the vacuum pump.

Danger to health when the vacuum pump is dismantled.

Danger to the environment.

Prior to shipping, the vacuum pump must imperatively be decontaminated and the degree of contamination must be documented in a declaration of decontamination ("Declaration of Decontamination"), which can be downloaded from www.buschvacuum.com.

Busch service will only accept vacuum pumps that come with a completely filled in and legally binding signed form.

Removal from service

Temporary removal from service

Prior to disconnecting inlet and outlet pipes as well as cooling water pipes, make sure that all piping is vented to atmospheric pressure

Recommissioning



CAUTION

After a long period of inactivity, it is possible that the rotor screws of the COBRA NC vacuum pump are stuck.

Turn the rotor screws manually.

- Make sure that all gaskets, plugs or adhesive tapes have been removed.
- Start the vacuum pump as described in the chapter "Installation and Commissioning"

Dismantling and Disposal of the vacuum pump



DANGER

In case the vacuum pump has conveyed gases that have been contaminated with harmful foreign material which are harmful to health, the oil and the condensates will also be contaminated with harmful foreign material.

These foreign materials can infiltrate the pores, recesses and other internal spaces of the vacuum pump.

Danger to health during dismantling of the vacuum pump.

Danger to the environment.

During dismantling of the vacuum pump protective equipment and clothing must be worn.

The vacuum pump must be decontaminated prior to disposal.

Prior to shipping, the vacuum pump must imperatively be decontaminated and the degree of contamination must be documented in a declaration of decontamination ("Declaration of Decontamination"), which can be downloaded from www.buschvacuum.com.

Used oil and condensates must be disposed of separately in compliance with applicable environmental regulations.

When the vacuum pump comes to the end of its life:

- it must be decontaminated



CAUTION

Only authorised personnel may carry out dismantling work on the vacuum pump. Before work begins, the operator of the vacuum pump must fill in a form or a "Declaration of Decontamination" that provides information on possible dangers and appropriate measures.

If this form has not been filled in completely and signed, the vacuum pump may not be dismantled.

- drain the oil
 - ◆ dispose of the used oil in compliance with applicable environmental regulations
- dismantle the vacuum pump



CAUTION

During dismantling of the vacuum pump protective equipment and clothing must be worn

- ◆ dispose of the vacuum pump as scrap metal
- dispose of the different components of the pump in compliance with applicable regulations

Troubleshooting



WARNING

Risk of electrical shock, risk of damage to equipment.

Electrical installation work must only be executed by qualified personnel that knows and observes the following regulations:

- IEC 364 or CENELEC HD 384 or DIN VDE 0100, respectively,
- IEC-Report 664 or DIN VDE 0110,
- BGV A2 (VBG 4) or equivalent national accident prevention regulation.



CAUTION

During operation the surface of the vacuum pump may reach temperatures of more than 90 °C.

Risk of burns!

Let the vacuum pump cool down prior to a required contact or wear heat protection gloves.

Problem	Possible Cause	Remedy
The vacuum pump does not reach the usual pressure	The vacuum system or suction line is not leak-tight	Check the hose or pipe connections for possible leak
The drive motor draws a too high current (compare with initial value after commissioning)	Process deposits on the pumping components (the most common cause)	Flush the pump carefully
Evacuation of the system takes too long	In case a mesh screen is installed on the suction connection: The mesh screen on the suction connection is partly clogged	Clean the mesh screen If cleaning is required too frequently install a filter upstream
	In case an inlet filter is installed on the suction connection: The filter on the suction connection is partly clogged	Clean or replace the inlet filter, respectively
	Partial clogging in the suction, discharge or pressure line	Remove the clogging
	Long suction, discharge or pressure line with too small diameter	Use larger diameter
	Internal parts worn or damaged	Repair the vacuum pump (Busch service)
The vacuum pump does not start	The drive motor is not supplied with the correct voltage or is overloaded	Supply the drive motor with the correct voltage
	The drive motor starter overload protection is too small or trip level is too low	Compare the trip level of the drive motor starter overload protection with the data on the nameplate Correct if necessary In case of high ambient temperature: Set the trip level of the drive motor starter overload protection 5 percent above the nominal drive motor current
	One of the fuses has blown	Check the fuses
	The connection cable is too long causing a voltage drop at the vacuum pump	Use sufficiently dimensioned cable
	The vacuum pump or the drive motor is blocked	Turn the screws rotors manually (PMR) or repair the vacuum pump (Busch service)
The vacuum pump is blocked	The drive motor is defective	Replace the drive motor (Busch service)
	Solid foreign matter has entered the vacuum pump	Repair the vacuum pump (Busch service) Make sure the suction line is equipped with a mesh screen If necessary additionally provide a filter

	Corrosion in the vacuum pump from remaining condensate	Repair the vacuum pump (Busch service) Check the process Observe the chapter "Installation and Commissioning, Operating Notes"
	The vacuum pump was run in the wrong direction	Repair the vacuum pump (Busch service) When connecting the vacuum pump make sure the vacuum pump will run in the correct direction (see "Installation")
	Condensate ran into the vacuum pump	Repair the vacuum pump (Busch service) Make sure no condensate will enter the vacuum pump, if necessary provide a drip leg and a drain cock Drain condensate continuously
	The recirculating pump is blocked	Remove the recirculating pump, clean if necessary
The vacuum pump starts, but labours or runs noisily or rattles The drive motor draws a too high current (compare with initial value after commissioning)	Connections in the drive motor terminal box are defective	Check the proper connection of the wires against the connection diagram
	Not all drive motor coils are properly connected	Tighten or replace loose connections
	The drive motor operates on two phases only	
	The vacuum pump runs in the wrong direction	Verification and rectification see "Installation and Commissioning", correct if necessary
	Standstill over several weeks or months	Let the vacuum pump run warm with inlet closed
	Improper oil quantity, unsuitable oil type	Use the proper quantity of the recommended oils (see "Oil", oil change see "Maintenance")
	No oil change over extended period of time	Perform oil change incl. flushing (see "Maintenance")
The vacuum pump runs very noisily	Foreign objects in the vacuum pump Stuck bearings	Repair the vacuum pump (Busch service)
	Defective bearings	Repair the vacuum pump (Busch service)
The vacuum pump runs very hot (The surface of the vacuum pump shall not exceed a temperature of 100°C)	Worn coupling element	Replace the coupling element (Busch service)
	Cooling water flow too low	Check the cooling water circuit and adjust the flow if necessary
	Ambient temperature too high	Observe the permitted ambient temperatures
	Temperature of the inlet gas too high	Observe the permitted temperatures for the inlet gas
	Oil level too low	Top up oil Check the oil filler plugs Check the mechanical seals (option)
	Oil burning by an overheating of the vacuum pump	Drain oil Fill in new oil (see "Maintenance")
	Mains frequency or voltage outside tolerance range	Provide a more stable power supply
	In case a mesh screen is installed on the suction connection: The mesh screen on the suction connection is partially clogged	Clean the mesh screen If cleaning is required too frequently install a filter upstream

	In case an inlet air filter is installed on the suction connection: The filter on the suction connection is partially clogged	Clean or replace the filter
	Partial clogging in the discharge line	Remove the clogging
	Long discharge or pressure line with too small diameter	Use larger diameter
The oil is black	Oil change intervals are too long The oil was overheated	Drain oil Fill in new oil (see "Maintenance")

Spare parts

NOTE: For COBRA pumps, please contact your local Busch representative for any ordering of spare parts or concerning the servicing of the vacuum pump.

When ordering spare parts or accessories, please always quote the type ("Type") and the serial no. ("S/N") of the vacuum pump. This will allow Busch service to check if the vacuum pump is compatible with a modified or improved part.

The exclusive use of genuine spare parts and consumables is a prerequisite for the proper function of the vacuum pump and for the granting of warranty, guarantee or goodwill.

Find the list of companies all over the world (by the time of the publication of these installation and operating instructions) on the rear cover page.

Find the up-to-date list of Busch companies and agencies all over the world on the internet at www.buschvacuum.com.

Oil type/ quantity

Oil type

Busch VE 101 oil

- Make sure that the oil type corresponds to specifications:

Denomination	Busch VE 101
ISO-VG	100
Base	Diester
Density [g/cm ³]	0.96
Kinematic viscosity at 40 °C	95
Kinematic viscosity at 100 °C	9,5
Flashpoint [°C]	255
Pourpoint [°C]	-30
Part no. 1 litre packaging	0831 000 099
Part no. 5 litres packaging	0831 000 100

Oil quantity

The quantity of oil specified in the following table is of informative nature only. Check the oil level with the help of the various oil sight glasses on the vacuum pump.

Quantity	Motor side (A)	Suction side (B)
NC 0630 C	1,5 L	0,5 L

Cooling liquid type/ quantity

Cooling liquid type

- Make sure that the cooling liquid type corresponds to specifications :

Specifications	C-25 (ready-to-use)	
	25 litres can	5 litres can
	part no. 0831 563 468	part no. 0831 563 469

Cooling liquid quantity

The quantity of cooling liquid specified in this instructions manual is of informative nature only. Respect the procedure of filling the cooling liquid.

Quantity	C-25 (ready-to-use)
NC 0630 C	approx. 29 litres



CAUTION

The proportion of pure glycol and water for the cooling liquid corresponds to a percentage of 40% pure glycol and 60% water. It is therefore not necessary to prepare the mix before use. But when using pure glycol, it is imperative to prepare the mix prior to filling the pump and to respect this proportion.

Technical data

Technical data			NC 0630 C
Nominal suction capacity	50 Hz 60 Hz	m ³ /h m ³ /h	630
Ultimate pressure	50 Hz 60 Hz	mbar	≤ 0,01
Nominal motor rating with discharge counter pressure between 0,10 and 0,20 bar	50 Hz 60 Hz	kW kW	15,0 17,0
Nominal motor speed	50 Hz 60 Hz	min ⁻¹ min ⁻¹	3000 3600
Noise level (EN ISO 2151)	50 Hz 60 Hz	dB(A) dB(A)	< 70 < 75
Ambient temperature		°C	+5 - +50
Maximal counter pressure at the discharge side		bar	0,2
Cooling water requirement		l/ min	12
Cooling water pressure		bar	1 - 6
Cooling water temperature		°C	+10 - +30
Weight	approx.	kg	550



EC Declaration of Conformity

NOTE: This Declaration of Conformity and the **CE**-mark affixed to the nameplate are valid for the vacuum pump within the Busch-scope of delivery. When this vacuum pump is integrated into a larger machinery the manufacturer of the larger machinery (this can be operator, too) must conduct the conformity assessment process for the larger machine, issue the Declaration of Conformity for it and affix the **CE**-mark.

We

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declare that the screw vacuum pumps **NC 0630 C**

in accordance with the European Directives

"Machinery" 2006/42/EC,

"Electromagnetic Compatibility" 2004/108/EC

"Restriction of the use of certain hazardous substances in electrical and electronic equipment" 2011/65/EU

have been designed and manufactured to the following specifications:

Standards	Title of the standard
Harmonised standards	
EN ISO 12100-1 EN ISO 12100-2	Safety of machinery - Basic concepts, general principles of design - Part 1 and 2
EN ISO 13857	Safety of machinery - Safety distances to prevent hazard zones being reached by upper and lower limbs
EN 1012-1 EN 1012-2	Compressors and vacuum pumps - Safety requirements - Part 1 and 2
EN 60204-1	Electrical equipment of machines - Part 1:
EN 61000-6-1 EN 61000-6-3	Electromagnetic compatibility (EMC) – Generic standards – Immunity and emission for residential, commercial and light-industrial environments; Part 1 and 3
EN 61000-6-2 EN 61000-6-4	Electromagnetic compatibility (EMC) – Generic standards – Immunity and emission standard for industrial environments; Part 2 and 4
National standards	
EN ISO 2151	Acoustics - Noise test code for compressors and vacuum pumps - Engineering method (grade 2)

Manufacturer	Mandatory within the EC	Person authorised to compile the technical file
		
Christian Hoffmann General director	Dr.-Ing. Karl Busch General director	Gerd Rohweder Product manager

Note

Note

Busch – All over the World in Industry

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